

Beyond Entropy

Outlier-Aware Admission for Open-World Test-Time Adaptation

Pattern Recognition — CSE 4835 | Final Presentation

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Outline

RECAP

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2. An illustrative TTA pipeline
3. Evolution & foundational methods
4. Challenges and the research gap
5. Our chosen direction: open-world TTA

THIS WORK

6. STAMP: the pipeline we build on
7. Four falsifiable interventions
8. Results & analysis
9. Conclusion & future work

Motivation

- Deployment data rarely matches training data

Distribution shift — mismatch between source and target distributions

- Accuracy degrades sharply once shift appears
- Retraining is often impossible: no source data, no labels, must react instantly



Train

Clean, uncorrupted source data

↓ *distribution shift*



Test

Corrupted *and* open-world:
unknown classes too

What Is Test-Time Adaptation?

“Adapting a pretrained model to unlabeled test data – on the fly, at deployment.”

TTA updates a source-trained model using only the unlabeled target stream, guided by self-supervised signals (entropy, pseudo-labels, feature statistics) – no retraining, no target labels.

CORE ASSUMPTIONS



**No Source
Data**



**No Target
Labels**



**Model Access
Only**



**Online &
Limited**

Example: CIFAR-10 $\rightarrow$ CIFAR-10-C



No labels used at any point — same protocol as CIFAR-10-C / CIFAR-100-C later.

Evolution & Foundational Methods

FROM TRANSFER LEARNING TO TTA

1. Transfer Learning

↓ drop target labels

2. Domain Adaptation

↓ drop source data

3. Source-Free DA

↓ go online / streaming

4. Test-Time Adaptation

APPLICATIONS

Driving, medical imaging, robotics, segmentation,
OOD detection, VLMs

KEY METHODS THIS WORK BUILDS ON



Tent (ICLR'21)

Entropy min. on BN params —
base loss



EATA (ICML'22)

Filters unreliable high-entropy
samples



SAR (ICLR'23)

Sharpness-aware — STAMP's
optimizer

Challenges → Research Gap

KNOWN CHALLENGES

- **Catastrophic forgetting** under continual updates
- **Confirmation bias** from noisy pseudo-labels
- **Unstable optimization** under severe shift
- **Warm-up statistics** are unreliable

THE OPEN-WORLD GAP

- Most TTA assumes a *closed world*: every sample is a known class
- Real streams contain **unknown-class outliers**
- Adapting on confident errors **reinforces mistakes**
- Joint recognition + rejection is **under-explored**

Our target: stay accurate *and* learn to reject what the model has never seen.

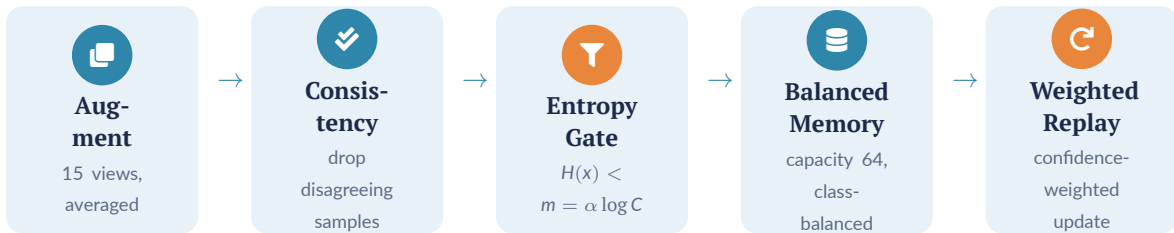
Our Chosen Direction



Open-world TTA: stay accurate **without** letting unknown samples corrupt the update.

- We adopt **STAMP** (Yu et al., ECCV 2024) — built for exactly this joint setting
- Its detector *and* its adaptation objective both rely on **prediction entropy** — one signal, two jobs
- **Research question:** can the detector be replaced by a signal from OOD detection, leaving the rest of STAMP untouched?

STAMP: The Pipeline We Build On



$$\mathcal{L}_{\text{STAMP}} = \frac{1}{|\mathcal{M}|} \sum_{x_i \in \mathcal{M}} \tilde{w}_i H(f_{\theta}(x_i)), \quad \tilde{w}_i = \frac{|\mathcal{M}| w_i}{\sum_j w_j}, \quad w_i = \exp[-H(f_{\theta}(x_i))]$$

⇒ Only normalization parameters are updated, via the SAM-style two-step update. **The catch:** entropy $H(x)$ decides *both* admission *and* the loss – coupling detection to a classifier whose confidence keeps changing.

Problem Formulation & Experimental Principle

Let f_θ be a source-trained classifier with logits $\mathbf{z}(x) = f_\theta(x)$. The test stream mixes:

- corrupted **in-distribution (ID)** samples — must still be classified correctly
- **out-of-distribution (OOD)** samples — scored as outliers, never used as adaptation targets

No test label is ever used for adaptation or threshold fitting — only for evaluation.

Guiding principle: change only the outlier score, hold the rest of STAMP fixed.

Any performance change is then attributable to the score alone.

Evaluation Protocol

- **Datasets:** CIFAR-10-C, CIFAR-100-C — severity 5, 15 corruptions
- **OOD mixture:** 10,000 ID + 2,500 noise outliers per corruption (80:20)
- **Backbone:** ResNet-18, batch 64, reset every corruption domain
- **Baselines:** source model, norm_test, modified STAMP
- **Metrics:** Accuracy (ID), AUC (ID vs. OOD), and the joint

$$H\text{-score} = \frac{2 \text{Acc} \cdot \text{AUC}}{\text{Acc} + \text{AUC}}$$

Roadmap: Four Falsifiable Interventions



R1

KNN feature
distance

done



R2

Logit energy
score

done



R3

Adaptive
conformal
threshold

running



R4

Replay-loss
scaling fix

done

- R1–R2 replace *what* scores a sample; R3 replaces *how* it becomes a decision
- R4 is not a new score — it's a correctness fix to STAMP's own replay loss
- Every round keeps the consistency filter, memory, and replay objective fixed

Round 1 — KNN Feature-Distance Score

Hypothesis: feature geometry catches outliers an entropy threshold misses, decoupled from the classifier's own confidence.

$$s_{\text{KNN}}(x) = 1 - \frac{1}{k} \sum_{j \in \mathcal{N}_k(x)} \hat{\mathbf{h}}(x)^\top \hat{\mathbf{r}}_j$$

- $\hat{\mathbf{h}}(x)$: normalized penultimate feature; $\mathcal{R}$: bank of clean source features
- Up to 2,000 references, $k=10$; replaces entropy for *both* admission and AUC
- Consistency gate, memory, and replay update stay unchanged

Expected: independence from softmax confidence $\Rightarrow$ better AUC. **Not what we observed.**

Round 1 – Results & Diagnosis

CIFAR-10-C	Acc.	AUC	H
Source	57.34	70.39	62.32
Norm-test	72.95	68.51	70.58
KNN-STAMP	78.64	59.10	63.61

CIFAR-100-C	Acc.	AUC	H
Source	35.83	43.13	38.01
Norm-test	45.90	80.91	58.49
KNN-STAMP	45.73	69.93	54.15

- KNN-STAMP **wins accuracy** (+21.3 pts over source) but **loses AUC** (−9.4 vs. norm-test on C10, −11.0 on C100)
- **Working explanation:** corruption shifts the clean reference bank away from ID features too — distance rejects *useful* samples, not just outliers

Round 2 — Energy Score with Online Calibration

Hypothesis: drop the fixed reference bank; score directly from logits.

$$E_T(x) = -T \log \sum_{c=1}^C \exp(z_c(x)/T), \quad T=1$$

- Uses logits STAMP already computes — **no extra forward pass, no reference bank**
- Threshold = **80th percentile** of the latest 1,000 scores (matches 80:20 mixture, label-free)
- First 256 samples fall back to entropy margin (warm-up); cutoff resets each domain

$$E_T(x) < \text{Quantile}_{0.80}(\mathcal{B}_t)$$

Round 2 — Results & Diagnosis

CIFAR-10-C	Acc.	AUC	H
Norm-test	72.95	68.51	70.58
KNN-STAMP	78.64	59.10	63.61
Energy-STAMP	78.99	54.33	63.80

CIFAR-100-C	Acc.	AUC	H
Norm-test	45.90	80.91	58.49
KNN-STAMP	45.73	69.93	54.15
Energy-STAMP	57.14	95.12	71.33

- **CIFAR-100-C:** strongest result yet — +17.2 H-score over KNN, +12.8 over norm-test
- **CIFAR-10-C:** AUC *drops* to 54.33 — energy alone is **not** dataset-independent

Round 3 — Adaptive Conformal Thresholding

running

Rounds 1–2 change the *score*; Round 3 asks: does **online threshold calibration** close the gap, independent of score choice?

$$\alpha_{t+1} = \text{clip}_{[0,1]} (\alpha_t + \gamma(\rho - r_t))$$

- $\rho = 0.20$: target rejection rate; r_t : observed rate in the current batch
- Cutoff read at quantile $1 - \alpha_t$ of the rolling buffer — rises/falls automatically

Entropy-STAMP + ACI
isolates the threshold mechanism

Energy-STAMP + ACI
tests if calibration rescues C10-C

Both variants mid-run; results reserved for the full table.

Round 4 — The Bug: Replay-Loss Scaling

Not a new score — an **implementation fix** in STAMP's own replay loss.

✘ Before (bug)

$$\mathcal{L}_{\text{old}} = \frac{64}{N} \mathcal{L}$$

Normalized for capacity 64, then
averaged over the *actual* size N

✔ After (fixed)

$$\tilde{w}_i = \frac{Nw_i}{\sum_j w_j}, \quad \mathcal{L} = \frac{1}{N} \sum_i \tilde{w}_i H_i$$

Coefficients always sum to N — a proper
weighted average

- Memory resets every corruption domain and starts **partially filled** ($N < 64$) $\Rightarrow$ gradients inflated up to **64×** right after every reset
- Full-memory case ($N=64$) is unchanged — only warm-up is rescaled. We name this fix **STAMP+BSN**.

Round 4 — Results: Best Completed Scores

CIFAR-10-C	Acc.	AUC	H
Norm-test	72.95	68.51	70.58
Energy-STAMP	78.99	54.33	63.80
STAMP+BSN	78.95	83.96	81.19

CIFAR-100-C	Acc.	AUC	H
Norm-test	45.90	80.91	58.49
Energy-STAMP	57.14	95.12	71.33
STAMP+BSN	58.02	98.29	72.89

- CIFAR-10-C AUC jumps **54.33** → **83.96** (+29.6 pts) with accuracy preserved — unstable warm-up gradients were destabilizing the model
- Correct replay-loss scaling matters as much as the choice of outlier score

Full Results: All Rounds, Both Benchmarks

Dataset	Method	Acc.	AUC	H-score
CIFAR-10-C + Noise	Source	57.34	70.39	62.32
	Norm-test	72.95	68.51	70.58
	KNN-STAMP (R1)	78.64	59.10	63.61
	Energy-STAMP (R2)	78.99	54.33	63.80
	STAMP+BSN (R4)	78.95	83.96	81.19
	Entropy+ACI (R3)		<i>running</i>	
	Energy+ACI (R3)		<i>running</i>	
CIFAR-100-C + Noise	Source	35.83	43.13	38.01
	Norm-test	45.90	80.91	58.49
	KNN-STAMP (R1)	45.73	69.93	54.15
	Energy-STAMP (R2)	57.14	95.12	71.33
	STAMP+BSN (R4)	58.02	98.29	72.89
	Entropy+ACI (R3)		<i>running</i>	
	Energy+ACI (R3)		<i>running</i>	

15 severity-5 corruptions, mean across streams. **Bold** = best completed value per dataset.

Key Takeaways

-  **KNN distance** improves accuracy but **worsens** detection — the clean reference bank is the wrong anchor once ID features drift.
-  **Energy scoring** excels on CIFAR-100-C (71.33 H) but is **dataset-dependent** — weak on CIFAR-10-C.
-  **The replay-loss fix (R4)** gives the **best results on both benchmarks** — a bug fix beat every scoring idea.
-  Can **threshold calibration (R3)** alone close the CIFAR-10-C gap? Still running.

Conclusion

We asked whether STAMP's entropy-based detector — which doubles as its adaptation objective — can be replaced by an OOD-detection signal, holding the rest of STAMP fixed.

- Neither KNN distance nor energy scoring alone gives a **dataset-independent** gain in the joint trade-off
- The biggest, most consistent gain came from a **correctness fix** (STAMP+BSN) — 81.19 H on CIFAR-10-C, 72.89 on CIFAR-100-C, best on both
- **Limitation:** R3 (ACI) is still running — we cannot yet isolate calibration from score choice on CIFAR-10-C

Future Work



Finish Round 3

Complete Entropy+ACI and Energy+ACI; report AUC, H-score.



Combine Best of Each Round

Energy score + BSN fix + ACI calibration together.



Per-Corruption Diagnostics

ID-rejection / OOD-admission rates per corruption type.



Warm-Up Ablation

Log memory occupancy and loss to confirm *why* STAMP+BSN stabilizes CIFAR-10-C.

References

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